EMOTION TRACKING VISUAL SCHEDULER

Design Expo • Spring 2022

PURDUE UNIVERSITY
College of Engineering

Team SangFroid
(Multidisciplinary Engineering)

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Problem Frame and Scope

Anxiety affects a large portion of the general population. For the purposes of this project, the solution focuses on children between the kindergarten and sixth grade classrooms with low- to mid-support Autism Spectrum Disorder (ASD). The device needs to be portable, so it is easily transported between classrooms and home. As a tool for integrated public classrooms, the device will not incorporate therapy, will be nonrestrictive, and will promote self-management. Lastly, the device must be modular so minor customizations can be made to attend to the specialized needs of everyone. This led us to develop the problem frame below:

How can we create a device that mitigates the impact of anxiety experienced by children with autism in the classroom as well as allow researchers and professionals to better understand behavioral responses?

Stakeholders and Users

Users	Teachers/Instructors
	Parents and Guardians
	K-6 Children with ASD
Professionals	Mental Health Professionals
	Local Education Organizations
	IEP Developers
Other Stakeholders	Product Developers
	School Management Software Companies

The Solution

A visual scheduling device that empowers children with ASD to feel in control of their daily activities, while also allowing teachers, paraprofessionals, and researchers to track basic emotional response of the child during said activities.

Figure 1 – Final Physical Prototype

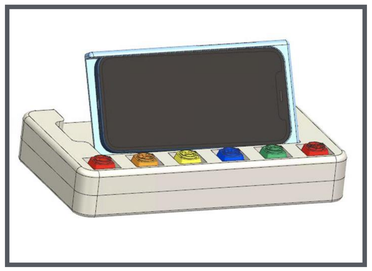

Figure 3 – Complete CAD Model

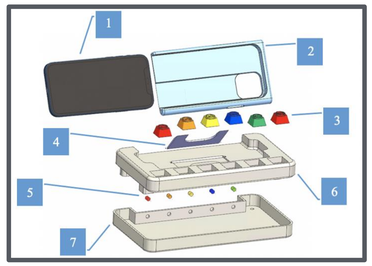

Figure 2 – Exploded CAD Model

Item Number	Quantity	Name
1	1	iPhone 13
2	1	Phone/Device Holder
3	6	Emotion Tracking Buttons
4	1	Kickstand
5	5	LED Lights
6	1	Base Top
7	1	Base Bottom

Figure 2.1 – Exploded CAD Model Legend

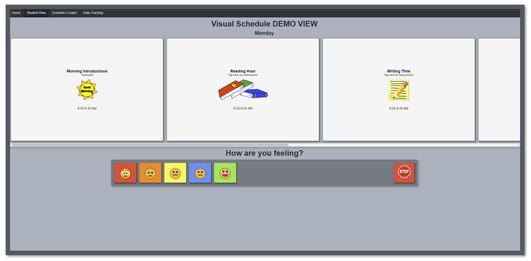

Figure 4 – Student View

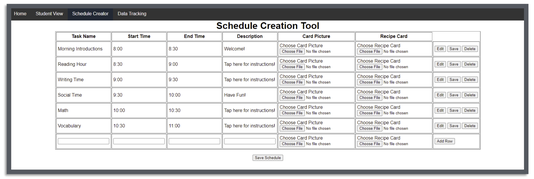

Figure 5 – Schedule Creation Tool

Solution Validation

- Increasing diagnoses of ASD (Coe, 2020)
- Common co-diagnoses of ASD and anxiety
- Success of integration of Five-Point Scale
- Success of 'Visual Stop' indicators
- Reduction of anxiety due to visual schedule (Lequia, et al., 2011)

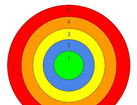

Figure 6 – Five Point Scale currently in use at TSC

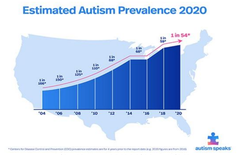

Figure 7 – Increasing Autism Diagnoses (Coe, 2020)

Stakeholder Feedback

"… I feel like overall its user friendly, for the student and para/teacher."
– Stacia Bernard, Director of Youth Services, Grant's House

"I think a child in the functioning range of which you are targeting could learn this pretty quickly. Especially if someone models the usage of it to them…"
– Alicia Pearson, MA, BCBA, Cornerstone Autism Center

"[Activity scheduling helps prevent uncertainty, provide structure, and often includes a visual break card to help indicate distress (such as a stop sign or red light).]" – Stacia Bernard, Director of Youth Services, Grant's House

Further Design Details

Physical Model

The CAD model was designed in Siemens NX and is divided into five unique parts: the bottom section of the base, the top section of the base, the phone case, the kickstand, and the buttons, see Figure 2.

Web App

The website was designed using a combination of HTML, CSS, and JavaScript. It was developed as a static, but interactive, demonstration of the intended platform. The website consists of a three primary sections, a student view, schedule creation tool, and a data collection view. This was developed in such a way that it can also exist as a stand-alone platform. In the case that a child may already possess assistive device, the platform alone is still available.

Electronics

The light and button system was created using an Arduino uno microcontroller, and a simple circuit containing five LEDs and resistors. This is designed as a simple way to demonstrate how the teachers would be alerted to the student's emotions.

Testing and Validation

Simulation

We conducted an extensive FEA to test the performance of the device in several possible scenarios. The example shown to right is meant to be a simulation of a child stomping on the face of the device.

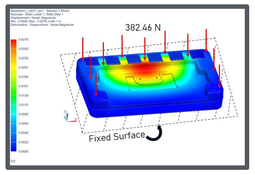

Figure 8 – Simulated Physical Response to a Child Stomping on the Device

Empirical

We conducted physical testing to ensure the structural integrity of the device. This testing included compression, drop, and simulated throwing tests. The diagram to the right demonstrates how the compression testing was conducted. The device was able to sustain a weight of 220 lbs. without plastic deformation occurring.

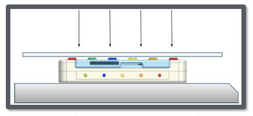

Figure 9 – Diagram of Compression Test

Future Steps

Physical Model

Going forward, we would like to redesign the hinge mechanism to be more fluid and operable. Additionally, we would like to redesign the electronics cavity to more precisely fit internal components.

Web App

For the web app, the primary desired change would be the integration of a backend database, such that the data from the child's use could be tracked, and so that the schedule creation tool could dynamically update the student's schedule. Additionally, infrastructure would be implemented to allow for multiple student's schedules to be controlled from a dashboard by the instructor/teacher..

System Integration

Currently, the electronics and web app are separate systems. In a final product, or next prototype, we would like to integrate the entirety of the system. This could be done through a method such as using a RaspberryPi, or through connecting the buttons to the phone through a BLE module.

Abbreviations

ASD - Autism Spectrum Disorder
ABA – Applied Behavioral Analysis
FEA – Finite Element Analysis
IEP – Individual Education Plan
PLA – Polyactic Acid
UI/UX – User Interface/User Experience

Acknowledgements

Many thanks to the professionals we worked with at Cornerstone Autism Center, Piece by Piece ABA Center, Tippecanoe County Schools, Greater Lafayette Area Special Services, Grant's House/Wabash Center, Purdue Professors Dr. Rispoli, Dr. Mason, and Dr. Hynes, and TA Andrew Gray.

References

Coe, M. (2020). *CDC estimate on autism prevalence increases by nearly 10 percent, to 1 in 54 children in the U.S.* Autism Speaks. Retrieved April 27, 2022, from https://www.autismspeaks.org/press-release/cdc-estimate-autism-prevalence-increases-nearly-10-percent-1-54-children-us

Lequia, J., Machalicek, W., & Rispoli, M. J. (2011, September 8). *Effects of activity schedules on challenging behavior exhibited in children with autism spectrum disorders: A systematic review*. Research in Autism Spectrum Disorders. Retrieved April 27, 2022, from https://www.sciencedirect.com/science/article/pii/S1750946711001334